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# BOSON GAS: BOSE-EINSTEIN STATISTICS AND BOSE-EINSTEIN CONDENSATION

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**Abstract:** When the atoms in a gas are cooled to extremely low temperatures, their quantum mechanical nature starts to become apparent on a macroscopic scale. Under the appropriate conditions, the atoms will ‘condense’ into a single quantum state of the system - a phenomenon known as Bose–Einstein condensation (BEC) where a large fraction of the bosons occupy the lowest quantum state of the external potential. This new state of matter has displayed many remarkable properties, stimulating continuous and exciting developments in the field of ultracold atoms since the first experimental realization of BEC in atomic gases in 1995. This review paper focused on the key theoretical concepts underlying the physics of Bose–Einstein statistics, thereby elucidating the phenomenon of BEC.

## 1. INTRODUCTION

The basic idea of Bose–Einstein condensation (BEC) dates back to 1925 when A. Einstein [11], on the basis of a paper of the Indian physicist S. N. Bose (1924) devoted to the statistical description of the quanta of light, predicted the occurrence of a phase transition in a gas of non-interacting atoms. This phase transition is associated with the condensation of atoms in the state of lowest energy and is the consequence of quantum statistical effects.

Although some characteristics of the phenomenon are evident in superfluidity in liquid helium and superconductivity in metals [9], it was not until 1995 that unambiguous experimental evidence of BEC was obtained. In a ground-breaking series of experiments, researchers at the JILA and MIT<sup>1</sup> were able to cool a dilute gas of alkali atoms to within a millionth of a degree of absolute zero, entering the regime where the quantum mechanical nature of the system dominates the behaviour of the gas. In these experiments, atomic gases are generated within vacuum chambers, laser light slows down the atoms, subsequently trapping and further cooling them to temperatures as low as a few hundred nanokelvin. Under specific conditions where the atoms behave as bosons and the temperature is significantly reduced, they undergo a phase transition, forming a Bose–Einstein condensate. This exotic state of matter, distinct from conventional solid or liquid states, can only be comprehended within the framework of quantum theory. In the Bose–Einstein state, all atoms occupy the same quantum state, namely the lowest energy state within the magnetic trap that confines them. Due to the large wavefunctions of individual atoms, they overlap. The phases of the waves are aligned across the entire cloud, effectively behaving as one collective wave. Despite its quantum-scale population, a BEC typically spans a few micrometers in diameter. Remarkably, the macroscopic wavefunction of a BEC becomes visually apparent when two condensates are al-

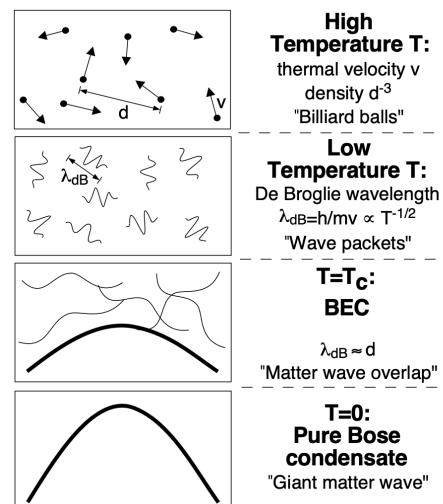


FIG. 1. Schematic representation illustrating the criterion for Bose–Einstein condensation: A weakly interacting gas behaves akin to a system of “billiard balls” at high temperatures. In a simplified quantum framework, the atoms are conceptualized as wavepackets with an extension denoted as  $\lambda_{dB}$ . At the transition temperature for BEC,  $\lambda_{dB}$  reaches a scale comparable to the inter-atomic distance, facilitating the formation of a Bose condensate. As the temperature tends toward absolute zero, the thermal cloud dissipates, yielding a pristine Bose condensate[7].

lowed to overlap, producing interference patterns akin to those observed in a double-slit experiment with light!

Bose–Einstein condensation is based on the fundamental concepts of quantum mechanics, specifically the indistinguishability and wave nature of particles. In a simplified representation, one can view atoms within a gas as quantum-mechanical wavepackets, possessing a size comparable to a thermal de Broglie wavelength  $\lambda_{dB} = (2\pi\hbar^2/mk_B T)^{1/2}$  where  $T$  is the temperature and  $m$  the mass of the atom.  $\lambda_{dB}$  can be regarded as the position uncertainty associated with the thermal momentum distribution[3]. The lower the temperature, the longer  $\lambda_{dB}$ . When atoms are cooled to the point where  $\lambda_{dB}$  is comparable to the interatomic separation, the atomic wavepackets “overlap” and the indistinguishability of particles becomes important as depicted in Fig.1. At this temperature, bosons undergo a phase transition and form a Bose–Einstein condensate. The transition temperature and the peak atomic density  $n$  are related as  $n\lambda_{dB}^3 \approx 2.612$  [6].

<sup>1</sup> The Nobel Prize in Physics 2001 was awarded jointly to Eric A. Cornell, Wolfgang Ketterle and Carl E. Wieman “for the achievement of Bose–Einstein condensation in dilute gases of alkali atoms, and for early fundamental studies of the properties of the condensates.” [1] At the time of the award, Cornell and Wieman were affiliated with the University of Colorado, Joint Institute for Laboratory Astrophysics (JILA), and Ketterle was affiliated with Massachusetts Institute of Technology (MIT).

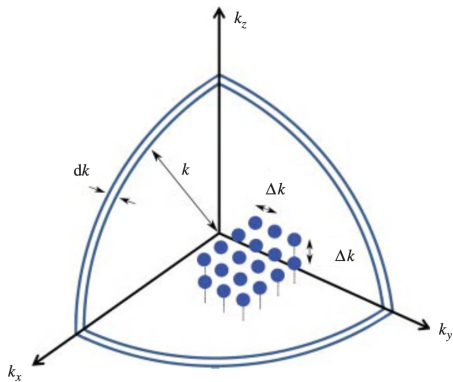


FIG. 2. The picture depicts a wavefunctions describing a single particle confined to a cubic volume  $V = l^3$  take the form of standing waves specified by wavevectors that form a cubic array in what is called **k**-space, with allowed components  $k_j = n_j \Delta k$  defined by positive integers  $n_j$ , some of which are shown. We divide this phase space of a single particle into macrostates with a magnitude of wavevector in the range  $k \rightarrow k + dk$  and estimate their microstate multiplicity by counting the number of single-particle microstates within the octant shell indicated [4].

## 2. BOSONS IN A SINGLE PARTICLE STATE

The single particle state may be visualized as a set of points arranged as a 3D cubic lattice, each specified by a wavevector  $\mathbf{k} = (k_x, k_y, k_z)$  with magnitude  $k$ . The points lie in what is called **k**-space, or sometimes reciprocal space, since in a manner of speaking it is an inverse of the space corresponding to particle position  $\mathbf{r}$ , in the sense that functions of  $\mathbf{r}$  can be represented as Fourier transforms of functions of  $\mathbf{k}$ , and vice versa. A point in this **r**-space represents a standing wave in **r**-space. From the quantisation conditions, the nearest neighbour distance between **k**-space lattice points is  $k = \pi/l$ . This is illustrated in Fig. 2.

The Pauli-Exclusion principle imposes no limit on the number of bosons that can occupy the same single particle energy level in a system of non-interacting particles[8]. The population of each energy level is controlled by the prevailing temperature through the Boltzmann factor and so at low temperature, multiple occupancy of some low-lying energy levels is expected to be significant. In these circumstances, we cannot employ the  $N$ -particle canonical partition function  $Z_N$  where one uses the one-particle partition function  $Z_1$  to construct a partition function  $Z_N^{dist} = Z_1^N$  for  $N$  distinguishable particles and then took approximate account of the indistinguishability by dividing by  $N!$ , a step that presumes that multiple occupancy of single particle states can be neglected.

Instead, the best way to explore the properties of the boson gas is through the grand canonical ensemble. Down the line, the statistical thermodynamic properties of a system characterized by a single quantised standing wave state in **k**-space is considered, and then deduced the prop-

erties of an entire collection of such single-particle states.

### 2.1. Grand Partition Function

The so-called *grand canonical ensemble* is an ensemble, each of whose members is itself a canonical ensemble with a fixed value for the number of particles,  $N$ [10]. In this way we allow the exchange of both energy and mass (via the number of particles) with a reservoir at temperature  $T$  and chemical potential  $\mu$ . All the statistical properties are to be derived from the grand partition function, which is of the form

$$Z_G(\mu, V, T) = \sum_i e^{-\beta(E_i - \mu N_i)} \quad (1)$$

where  $\beta = (K_B T)^{-1}$  and the occupation numbers  $n_i$  are subject to the condition  $\sum_i n_i = N$ . For a Bose gas  $n_i = 0, 1, 2, \dots$ . The sum is over all microstates of the system in a volume  $V$ , each characterised by an energy  $E_i$  and a number of particles  $N_i$ .

For a system consisting of a single particle state at energy  $\epsilon$ , the energy of a microstate is  $E_i = \sum_i \epsilon_i n_i$ , such that each is fully specified by the number of particles held. In this case, we get

$$Z_G^\epsilon(\mu, V, T) = \sum_{N_i=0}^{\infty} e^{\beta(\mu - E_i)N_i} = \frac{1}{1 - e^{\beta(\mu - \epsilon)}} \quad (2)$$

upon evaluating the sum as a geometric series. The superscript indicates that this is the grand partition function of the single particle state at energy  $\epsilon$ . The reservoir that provides energy and particles is essentially the remainder of the boson gas, distributed across all the other single particle states in the container, and presumed to act as a heat and particle bath with fixed temperature and chemical potential. Equivalently, we can instead imagine a more abstract source of energy and particles.

### 2.2. Bose-Einstein Statistics

The mean population in the single particle state at energy  $\epsilon$ , also known as the *occupation number*, is

$$\langle N \rangle_\epsilon = \frac{1}{\beta} \left( \frac{\partial \ln Z_G^\epsilon}{\partial \mu} \right)_{T, V} = \frac{1}{e^{\beta(\epsilon - \mu)} - 1} \quad (3)$$

using  $Z_G^\epsilon$  from Eq.(2). This expression is known as **Bose-Einstein Statistics**. It has several interesting features.

If the chemical potential of the reservoir increases, while its temperature remains constant, the mean population in the single particle state rises; explicitly

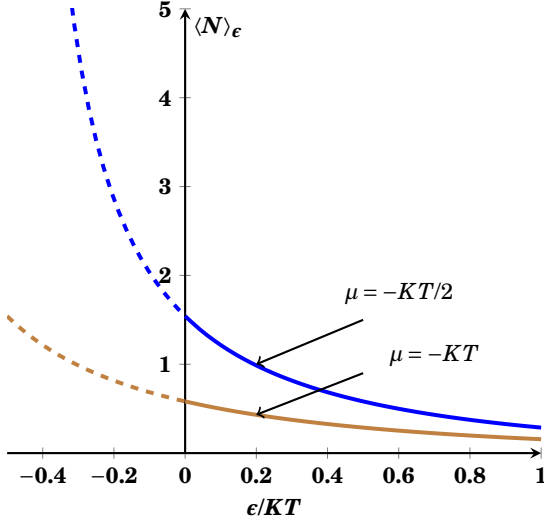


FIG. 3. Bose–Einstein occupation number of states with (positive) energy  $\epsilon$ , illustrated for two (negative) values of reservoir chemical potential  $\mu$  and a fixed temperature  $T$ . The dashed lines are continuations of  $\langle N \rangle_\epsilon$  for unphysical negative values of  $\epsilon$ , to illustrate the singularity at  $\epsilon = \mu$ . A figure produced based on Eq.(3).

$$\left( \frac{\partial \langle N \rangle_\epsilon}{\partial \mu} \right)_{T,V} = \beta \frac{e^{\beta(\epsilon-\mu)}}{(e^{\beta(\epsilon-\mu)} - 1)^2} > 0. \quad (4)$$

This restriction makes physical sense. However, it's crucial to note that the reservoir's chemical potential cannot surpass the energy level of state  $\epsilon$  to ensure that the denominator in Equation (3) remains non-negative, as the mean population must always be positive. As the chemical potential approaches  $\epsilon$  from below, the mean population can substantially increase, a scenario permissible for bosons.

In relation to the standing wave single particle states, which are treated as independent systems in contact with the same particle reservoir, they have energies (ref. Fig.2)

$$E_k = \frac{h^2}{8ml^2}(n_x^2 + n_y^2 + n_z^2) \quad (5)$$

varying from  $3h^2/(8mV^{2/3})$  (which is very small when  $V$  is large) up to infinity. We conclude that the chemical potential of the particle reservoir has to be less than  $3h^2/(8mV^{2/3})$ , or roughly speaking never positive. The implications of this strange assertion are discussed later.

Examining Eq.(3) with  $\mu < \epsilon$ , the mean populations in the single particle states decrease as the state energy  $\epsilon$  increases, for a given  $\mu$  and  $T$ , as illustrated in Fig.3. The singularity in the Bose–Einstein statistics expression lies safely in the unphysical region of negative  $\epsilon$ .

Another notable feature is that as the temperature of the reservoir increases while maintaining a constant

chemical potential, the mean population in the state once again rises i.e.,

$$\left( \frac{\partial \langle N \rangle_\epsilon}{\partial \mu} \right)_{\mu,V} = \frac{\epsilon - \mu}{KT^2} \frac{e^{\beta(\epsilon-\mu)}}{(e^{\beta(\epsilon-\mu)} - 1)^2} > 0 \quad (6)$$

We concluded that the mean populations in a set of single-particle states within a box can be controlled by adjusting the external reservoir parameters  $\mu$  and  $T$ . The sum of the occupation numbers of all the single-particle states, divided by the volume of the box, corresponds to the mean particle density  $n = \sum_\epsilon \langle N \rangle_\epsilon / V$ . It can be assumed that the chemical potential of such a gas will match with that of the reservoir since they will be in equilibrium.

It is quite illuminating to insert the expression

$$\mu = KT \ln \left( \frac{n}{n_q} \right) \quad (7)$$

for the reservoir chemical potential appropriate to the classical gas regime (with spin zero particles) with  $n \ll n_q$ , into Eq.(3). Note that  $n_q$  is the quantum concentration defined as

$$n_q = \frac{1}{\lambda_{dB}^3} \quad (8)$$

where  $\lambda_{dB}$  is the thermal de Broglie wavelength [7] expressed as

$$\lambda_{dB} = \left( \frac{h^2}{2\pi mKT} \right)^{1/2} \quad (9)$$

For  $\mu\beta \ll 0 < \beta\epsilon$  such that  $e^{\beta(\epsilon-\mu)} \gg 1$  we get

$$\langle N \rangle_\epsilon \approx e^{\beta(\mu-\epsilon)} = \frac{n}{n_q} e^{-\beta\epsilon} \ll 1 \quad (10)$$

attesting that the likelihood of multiple occupancy of any of the single particle states is very small if the density is much less than the quantum concentration  $n_q$ . Quantum gas thermodynamic properties emerge when the reservoir parameters are such that the mean density of particles in the system exceeds  $n_q$ . Eq.(7) and Fig.3 suggest that this would involve the approach of  $\mu$  towards zero from below, such that the single particle states with lowest energy become multiply occupied. But the relationship between  $\mu$  and  $n$  will then no longer be given by the classical expression, and the gas will deviate from classical behaviour, which is explored in the next sections.

### 3. THERMAL PROPERTIES OF A BOSON GAS

So far, the picture we have established is a single particle states forming a cubic array in  $\mathbf{k}$ -space, and

in equilibrium with an external reservoir following the Bose–Einstein statistics. The grand partition function for the gas as a product of grand partition functions for each of the single particle states labelled by  $\mathbf{k}_i$ :

$$Z_G(\mu, V, T) = \sum_{N_{\mathbf{k}_1}=0}^{\infty} e^{-N_{\mathbf{k}_1}\beta(\epsilon_{\mathbf{k}_1}-\mu)} \cdot \sum_{N_{\mathbf{k}_2}=0}^{\infty} e^{-N_{\mathbf{k}_2}\beta(\epsilon_{\mathbf{k}_2}-\mu)} \dots \quad (11)$$

The microstates are labelled by the set of populations  $N_{\mathbf{k}_i}$  across the array. This avoids assigning labels to any of the particles such that we can be sure that the result will be appropriate for indistinguishable particles. The grand partition function of the gas is therefore

$$Z_G = \prod_{\mathbf{k}_i} Z_G^{\epsilon_{\mathbf{k}_i}} = \prod_{\mathbf{k}_i} \frac{1}{1 - e^{\beta(\mu - \epsilon_{\mathbf{k}_i})}} \quad (12)$$

This is equivalent to  $\ln Z_G = \sum_{\mathbf{k}_i} \ln Z_G^{\epsilon_{\mathbf{k}_i}}$ , which can be written in the form  $\ln Z_G = \sum_{\epsilon} \Omega(\epsilon) \ln Z_G^{\epsilon}$  where the sum is over the energies of single particle states and  $\Omega(\epsilon)$  is the multiplicity of such states at a given energy  $\epsilon$ . As the single particle energies lie very close together, we might consider representing the sum as an integral over a continuous single particle energy  $\epsilon$ , denoting the multiplicity of single particle states in the range  $\epsilon \rightarrow \epsilon + d\epsilon$  as  $g(\epsilon)d\epsilon$ , where  $g(\epsilon)$  is a density of states expressed as

$$g(\epsilon) = \frac{(2s+1)V}{(2\pi)^2} \left( \frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \epsilon^{\frac{1}{2}} \quad (13)$$

and  $(2s+1)$  is the degeneracy of each state for particles of spin  $s$ , corresponding to different values of the quantum number  $m_s$ . We write

$$\ln Z_G \approx \int_0^{\infty} g(\epsilon) \ln Z_G^{\epsilon} d\epsilon = - \int_0^{\infty} g(\epsilon) \ln(1 - e^{\beta(\mu - \epsilon)}) d\epsilon \quad (14)$$

The average number of particles in the  $g(\epsilon)d\epsilon$  single particle states within the energy range  $\epsilon \rightarrow \epsilon + d\epsilon$  is equal to  $\langle N \rangle_{\epsilon} g(\epsilon)d\epsilon$ , with  $\langle N \rangle_{\epsilon}$  given by (3). Thus the mean number of particles in the system, across the entire set of single particle energy states, is

$$\langle N \rangle = \sum_{\mathbf{k}_i} \langle N \rangle_{\epsilon_{\mathbf{k}_i}} = \sum_{\epsilon} \Omega(\epsilon) \langle N \rangle_{\epsilon} \approx \int_0^{\infty} g(\epsilon) \langle N \rangle_{\epsilon} d\epsilon$$

This can be obtained more formally from (14) using

$$\begin{aligned} \langle N \rangle &= KT \left( \frac{\partial \ln Z_G}{\partial \mu} \right)_{T,V} \\ &= KT \int_0^{\infty} g(\epsilon) \left( \frac{\partial \ln Z_G^{\epsilon}}{\partial \mu} \right)_{T,V} d\epsilon \\ &= \int_0^{\infty} g(\epsilon) \langle N \rangle_{\epsilon} d\epsilon \end{aligned} \quad (15)$$

Similarly, from the definition of  $Z_G$  in (1), we can write

$$\begin{aligned} \langle E - \mu N \rangle &= KT^2 \left( \frac{\partial \ln Z_G}{\partial T} \right)_{\mu,V} \\ &= KT^2 \int_0^{\infty} g(\epsilon) \left( \frac{\partial \ln Z_G^{\epsilon}}{\partial T} \right)_{\mu,V} d\epsilon \\ &= \int_0^{\infty} g(\epsilon) (\epsilon - \mu) \langle N \rangle_{\epsilon} d\epsilon \end{aligned} \quad (16)$$

Comparing (15) and (16), this is consistent with our requirement that

$$\langle E \rangle = \sum_{\mathbf{k}_i} \epsilon_{\mathbf{k}_i} \langle N \rangle_{\epsilon_{\mathbf{k}_i}} = \sum_{\epsilon} \Omega(\epsilon) \epsilon \langle N \rangle_{\epsilon} \approx \int_0^{\infty} g(\epsilon) \epsilon \langle N \rangle_{\epsilon} d\epsilon \quad (17)$$

on the basis of the intuition that the total mean energy is the sum of mean energies  $\epsilon \langle N \rangle_{\epsilon} d\epsilon$  for each single particle state. Substituting (14) and (3) into (15) we then obtain

$$n(\mu, T) = \frac{\langle N \rangle}{V} = \frac{2s+1}{(2\pi)^2} \left( \frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \int_0^{\infty} \frac{\epsilon^{\frac{1}{2}} d\epsilon}{e^{\beta(\epsilon - \mu)} - 1} \quad (18)$$

for the mean density of the gas at a given chemical potential and temperature. If this could be inverted, it would give us the chemical potential  $\mu(n, T)$  of the gas in terms of the mean number of particles in the system at a given temperature, namely a replacement for the classical expression  $\mu = KT \ln[n/n_q(T)]$  appropriate for spin zero particles. However, this is not straightforward. Nevertheless the classical limit can be recovered under the appropriate assumption  $\epsilon - \mu \gg KT$ , or  $\mu \ll 0$ , and  $s = 0$ , such that

$$n \approx \frac{1}{(2\pi)^2} \left( \frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \int_0^{\infty} \epsilon^{\frac{1}{2}} e^{\beta(\mu - \epsilon)} d\epsilon = n_q e^{\beta\mu}, \quad (19)$$

where  $\mu \approx KT \ln(n/n_q)$  as before.

The mean energy of the gas per unit volume, for a given  $\mu$  and  $T$ , is written

$$e(\mu, T) = \frac{\langle E \rangle}{V} = \frac{2s+1}{(2\pi)^2} \left( \frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \int_0^{\infty} \frac{\epsilon^{\frac{3}{2}} d\epsilon}{e^{\beta(\epsilon - \mu)} - 1} \quad (20)$$

and in the classical limit with  $s = 0$ , this specific energy becomes

$$e(\mu, T) \approx \frac{1}{(2\pi)^2} \left( \frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \int_0^{\infty} \epsilon^{\frac{3}{2}} e^{\beta(\mu - \epsilon)} d\epsilon = \frac{3}{2} nKT, \quad (21)$$

just as we would expect for an ideal gas.

The Gibbs entropy of the gas is obtained from the grand partition function through the relation

$$TS_G = \langle E \rangle - \mu \langle N \rangle + KT \ln Z_G \quad (22)$$

such that according to (16) and (14) we get

$$\begin{aligned} S_G &= \frac{1}{T} \int_0^\infty g(\epsilon)(\epsilon - \mu) \langle N \rangle_\epsilon d\epsilon + K \int_0^\infty g(\epsilon) \ln Z_G^\epsilon d\epsilon \\ &= \frac{1}{T} \int_0^\infty g(\epsilon) \left[ \frac{\epsilon - \mu}{e^{\beta(\epsilon - \mu)} - 1} - KT \ln(1 - e^{\beta(\mu - \epsilon)}) \right] d\epsilon, \end{aligned} \quad (23)$$

which in the classical limit where  $\epsilon - \mu \gg KT$  can be shown to reduce to an entropy per unit volume  $s = S_G/V$  that goes to  $s_{cl} = nk(5/2 - \beta\mu)$ , or

$$s_{cl} = nk \ln \left( \frac{e^{5/2} n_q(T)}{n} \right) = \frac{3}{2} nk \ln \left( \frac{e^{5/3} T}{T_q(n)} \right) \quad (24)$$

which is the Sackur-Tetrode expression where

$$T_q(n) = \frac{h^2 n^{2/3}}{2\pi m K} \quad (25)$$

Finally, we consider the pressure of the gas. This is straightforward to obtain from the grand partition function when we write

$$p = KT \left( \frac{\partial \ln Z_G}{\partial V} \right)_{T, \mu} \quad (26)$$

From (14) the only  $V$ -dependence in  $\ln Z_G$  arises from the factor of  $V$  in the density of single particle states  $g(\epsilon)$  defined in (13), so

$$\begin{aligned} p &= \frac{KT}{V} \ln Z_G = \frac{KT}{V} \int_0^\infty g(\epsilon) \ln Z_G^\epsilon d\epsilon \\ &= -\frac{2s+1}{(2\pi)^2} \left( \frac{2m}{\hbar^2} \right)^{3/2} KT \int_0^\infty \epsilon^{1/2} \ln(1 - e^{\beta(\mu - \epsilon)}) d\epsilon \end{aligned} \quad (27)$$

which has classical limit  $p \rightarrow nkT$ .

At the outset, these expressions for the principal thermodynamic properties of the gas may seem daunting, but they provide insight when we choose a chemical potential and temperature and evaluate the associated particle, energy and entropy densities  $n, e, s$  and the pressure  $p$ . An example is given in Fig. 4, where the temperature is fixed and  $\mu$  is varied. Classical behaviour is recovered when  $\epsilon - \mu \gg KT$  such that  $e \rightarrow (3/2)nkT$  and  $p \rightarrow nkT$  while the entropy density tends towards the form  $s_{cl} = nk(5/2 - \mu/kT)$ . All these quantities start to deviate from classical behaviour when the particle density approaches and exceeds the quantum concentration  $n_q$ .

Nonclassical behaviour can also be revealed by eliminating  $\mu$  between  $e(\mu, V, T)$  and  $n(\mu, V, T)$ , for example, to give the energy density  $e(n, V, T)$  in terms of more familiar variables. As an illustration, we plot  $e$  and  $s$  against  $T$  at constant  $n$  for spin zero particles, and normalise the results by the classical expressions  $e_{cl} = (3/2)nkT$  and  $s_{cl} = nk \ln(e^{5/2} n_q/n)$ . To a certain extent, the behaviour

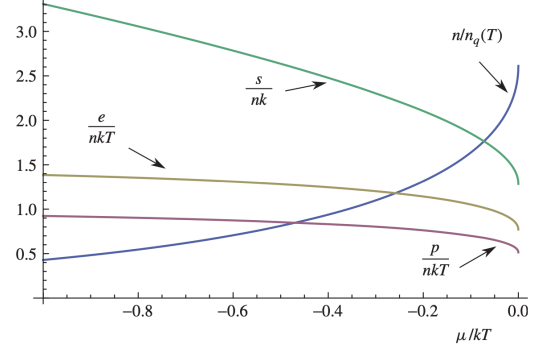


FIG. 4. The figure showing the properties of a gas of spin zero bosons as a function of the chemical potential at constant temperature. As  $\mu$  approaches zero from below, the particle density increases, and it exceeds the quantum concentration for  $\mu \gtrsim -0.4kT$ . The energy per particle begins to deviate from the classical value  $(3/2)kT$ , the pressure falls below  $nkT$  and the entropy per particle deviates from the classical result  $K(5/2 - \mu/kT)$  [4].

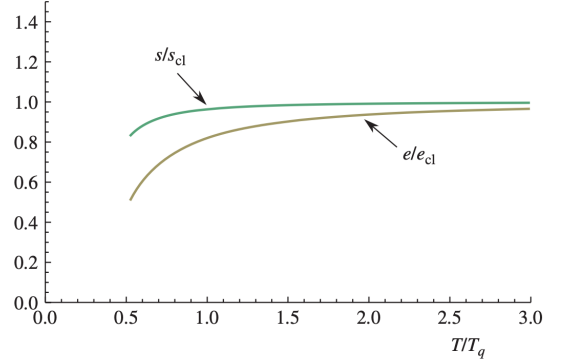


FIG. 5. The picture depicts the energy and entropy of a gas of spin zero bosons, per unit volume, divided by the classical expressions, as a function of temperature, illustrating quantum gas behaviour when  $T \lesssim T_q$  [4].

looks familiar. The classical mean energy is not appropriate for temperatures below  $T_q = h^2 n^{2/3} / (2\pi m k)$  which is the threshold temperature for quantum effects. The entropy is also reduced, which brings to mind the freezing out of degrees of freedom in the low temperature thermal behaviour of a diatomic gas, or of the Einstein solid, though the reasons for the deviation here are rather different. However, these expressions harbour a very serious problem, and simply cannot be entirely correct. This is partly illustrated by the fact that the energy and entropy are not reported below a temperature  $T \approx 0.5T_q$  in Figure 5. We investigate this problem in the next section.

#### 4. BOSE-EINSTEIN CONDENSATION

The nonclassical modifications to the thermal properties of a boson gas, observed in the previous section as the temperature decreases or the chemical potential increases, may not seem dramatic enough to account for the peculiar phenomena at low temperatures. However, a flaw lies hidden within Eq.(18), and by investigating it, we may uncover a rationale for such behavior.

The density of particles in the gas derived in (18) may be rewritten as

$$n(\mu, T) = \frac{2(2s+1)}{\pi^{\frac{1}{2}}} n_q \int_0^\infty \frac{x^{\frac{1}{2}} dx}{e^{-\beta\mu} e^x - 1}, \quad (28)$$

showing that the density increases as the (negative) chemical potential increases; explicitly

$$\left( \frac{\partial n}{\partial \mu} \right)_{T,V} = \frac{2(2s+1)}{KT\pi^{\frac{1}{2}}} n_q \int_0^\infty \frac{e^{-\beta\mu} e^x x^{\frac{1}{2}}}{(e^{-\beta\mu} e^x - 1)^2} dx > 0. \quad (29)$$

The problem that we concluded in section 2.2 that the chemical potential has an upper limit of (approximately) zero, such that the integral in (28) has an upper limit of  $\int_0^\infty x^{1/2}(e^x - 1)^{-1} dx \approx 1.306\pi^{1/2}$ . For lower (i.e. negative)  $\mu$ , the denominator in the integrand is larger than  $(e^x - 1)$  for all  $x \geq 0$  and the integral is therefore smaller. This suggests that the mean particle density of the system must satisfy

$$n \leq n_{max}(T) = 2.612(2s+1)n_q(T), \quad (30)$$

and this upper limit is apparent in the plot of  $n/n_q$  for spin zero particles in Fig.4. Does this mean that there is a temperature-dependent maximum number of particles that can be accommodated in the box? What happens if add more particles? More importantly, what happens if we maintain a certain particle density  $n$ , cease their exchange with the reservoir, and decrease the temperature until  $n_{max}(T)$  becomes lower than  $n$ , or equivalently when  $T \lesssim 0.5T_q$  approaches around half of  $T_q$  as depicted in Fig. 5? The particles cannot simply vanish!

Fortunately, the flaw in the development is easily identified. The approximation Eq.(15) is the root of the problem. An integral representation of the sum of mean populations over all the single particle states is an inadequate treatment for the very lowest states. The square root density of states (13) is a poor approximation at low  $\epsilon$ . It was derived by counting standing wave states in a thin shell in  $\mathbf{k}$ -space, on the assumption that there are many such states to count and the granularity of the lattice of allowed states did not matter. But as the gas approaches the quantum regime, the mean populations of particles in the low energy states become significant and an approximate treatment will not do.

In order to remove this deficiency, we recognise that the integral of populations over the model density of states accounts for only some of the particles. The populations in the lowest energy states should be represented explicitly. In the simplest treatment, we consider just the population in the lowest energy state  $\epsilon = 3h^2/8mV^{2/3} \approx 0$ . This solves the problem of the apparent disappearance of particles when  $T$  falls below  $0.5T_q$ : the particles that seemed not to find a home are actually accommodated in this state. The mean population in the single particle ground state at  $\epsilon \approx 0$  is therefore given by

$$\langle N \rangle_0 = \langle N \rangle - V n_{max}(T), \quad (31)$$

for room temperatures where  $n_{max}(T) < n$ , a condition that corresponds to

$$\begin{aligned} T &< \left( \frac{n}{2.612(2s+1)} \right)^{\frac{2}{3}} \frac{h^2}{2\pi m K} \\ &= \left( \frac{n}{2.612(2s+1)} \right)^{\frac{2}{3}} T_q = T_C, \end{aligned} \quad (32)$$

defining a temperature  $T_C$ , approximately equal to  $0.5T_q$  for a spin of zero. Since  $n_{max} \propto n_q \propto T^{3/2}$ , the relative proportion of particles occupying the ground state is given by

$$\frac{\langle N \rangle_0}{\langle N \rangle} = 1 - \frac{n_{max}(T)}{n} = 1 - \left( \frac{T}{T_C} \right)^{\frac{3}{2}}, \quad (33)$$

for  $T \leq T_C$ . The ground state appears to accumulate enormous numbers of particles when the temperature is decreased below  $T_C$ , and as  $T \rightarrow 0$ , all the particles fall into that state. The fractional occupation of the ground state as a function of temperature is illustrated in Fig.6, together with the proportion of particles that are not in the ground state, written  $\langle N \rangle_{\neq 0} / \langle N \rangle$ . This is multiple occupancy taken to an extraordinary degree.

The model now makes perfect sense. At a temperature of absolute zero, it satisfies our expectation that the particles should take the lowest available energy level. This would also be consistent with the third law: because there is just a single microstate available in this limit, the entropy of the gas will be zero. However, what is perhaps not expected is that there is a distinct temperature below which a gross ‘condensation’ of particles into the lowest energy single particle state begins. The word condensation suggests a phase transition of some kind, and indeed it is rather like that: the gas switches rapidly from a phase with properties that are a distorted version of those of a classical gas, to a phase that is altogether different. The phenomenon of the mass occupation of the lowest energy single particle state of the system is known as **Bose-Einstein condensation** and  $T_C$  is called the **Bose-Einstein condensation temperature**.

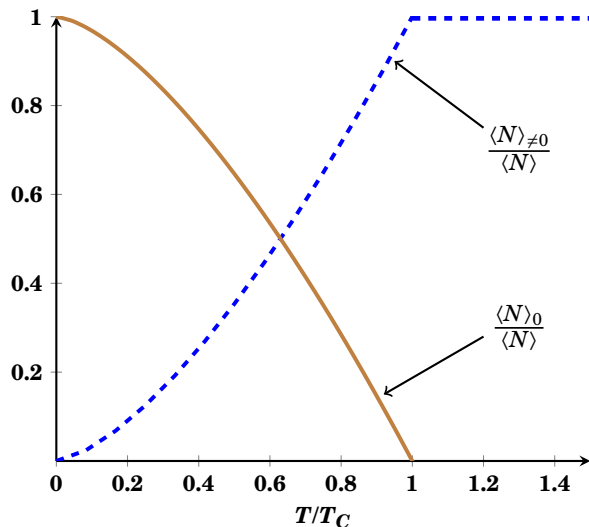


FIG. 6. The average number of particles of a boson gas occupying the ground state  $\langle N \rangle_0$  and occupying all other states  $\langle N \rangle_{\neq 0}$ , relative to the total population, varies with temperature.  $T_C$  represents the Bose–Einstein condensation temperature. A figure reproduced from [4] using Eq.(33).

We can illustrate the change in behaviour by sketching the entropy of the gas as a function of temperature at constant density in Fig.7. The entropy takes its classical logarithmic dependence on  $T$  for  $T \gg T_C$ , but this cannot remain valid as the temperature is reduced since it would imply that the entropy eventually became negative. Instead, from Fig.4 we see that the entropy falls to approximately  $1.3k$  per particle at  $\mu \approx 0$ , or equivalently at  $T = T_C$ , and then the entropy continues towards zero as the temperature is reduced below this threshold in proportion to the number of particles  $\langle N \rangle_{\neq 0}$  that do not lie in the single particle ground state. The population in the ground state  $\langle N \rangle_0$  contributes zero entropy. It is as though the gas separates into two components with very different thermodynamic properties.

Fig.8 illustrates how the chemical potential is logarithmic in temperature in the classical regime  $T \gg T_C$ , taking the form  $\mu \approx -(3/2)k \ln(T/T_q)$ , but approaches zero asymptotically from below for  $T < T_C$ . Further implications are that the energy of the gas is proportional to  $T$  in the classical regime, but is considerably suppressed for temperatures below  $T_C$  as a result of particle condensation into the  $\epsilon \approx 0$  ground state. The gas pressure is also suppressed.

So what can we say about a gas in which a considerable fraction of the particles reside in the lowest standing wave quantum mechanical state? The first thing we must abandon is the classical picture of particles moving around in a chaotic fashion and colliding with one another. The picture provided by quantum mechanics is one where particles are moving (slowly!) backwards and forwards between the walls and never seeming to interact with each other. It is hard to get a feel for such a gas, but

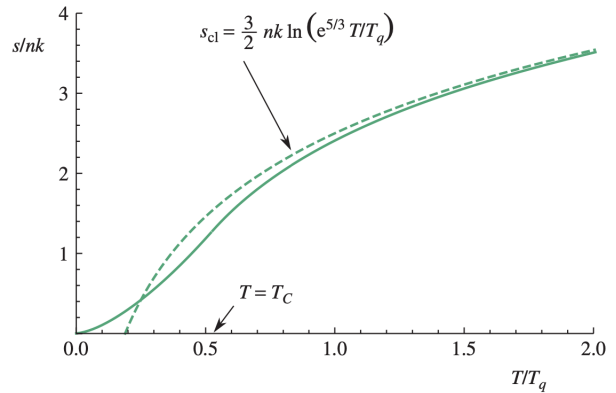


FIG. 7. Entropy per particle  $s/n$  for a gas of spin zero bosons as a function of temperature at constant density, illustrating its deviation from the classical Sackur–Tetrode expression (24) as the temperature falls below the Bose–Einstein condensation temperature  $T_C$ . [4].

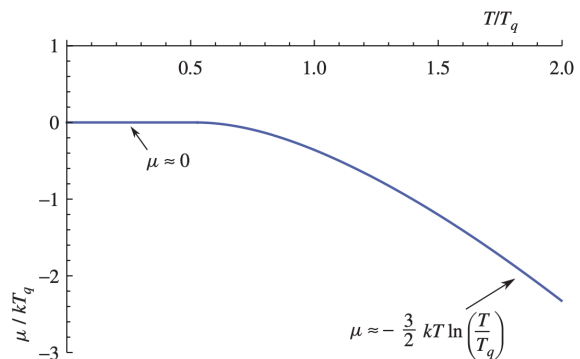


FIG. 8. Temperature dependence of the chemical potential of a gas of spin zero bosons, at a fixed density, demonstrating the change in behaviour at the Bose–Einstein condensation temperature  $T_C \approx 0.5T_q$ . [4].

there ought to be at least two clear signatures: firstly that the distribution of particle speed is very narrow, as most of the particles are occupying the same energy level, and secondly that the viscosity of such a fluid ought to be very low, as the particles rarely transfer momentum to each other. Resistance to the shearing of a fluid fundamentally arises from collision processes on a microscopic scale that spread out the effects of an external impulse.

#### 4.1. A Slight Touch on the Experimental Realization of BEC in 1995

Note that composite particles, such as atoms, also behave as bosons or fermions. Depending on the number of electrons, protons and neutrons, an atom can have integer or half-integer total spin and, therefore, be a boson

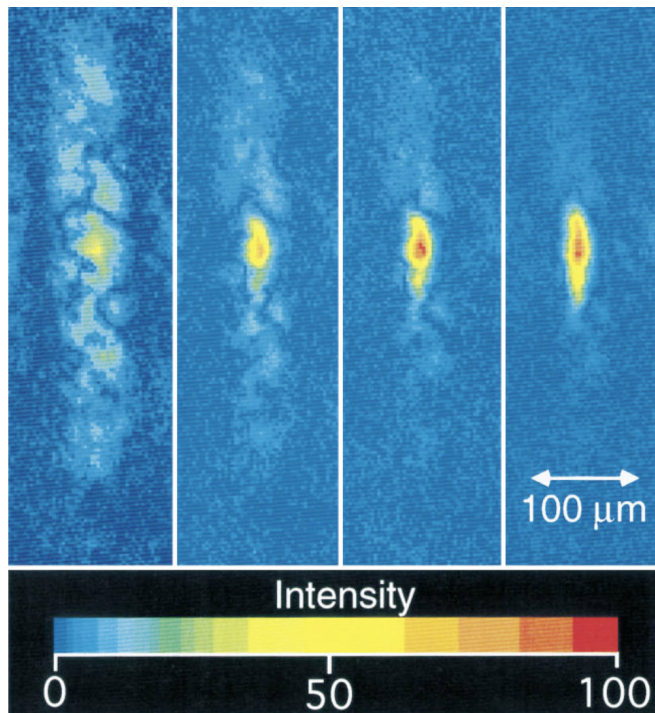


FIG. 9. Direct observation of BEC of magnetically trapped sodium atoms by dispersive light scattering. The clouds have condensate fractions that increase from close to 0% (left) to almost 100% (right). The dense core is the condensate, and the more diffuse cloud is the normal component. The signal for the normal component is rather weak and appears patchy due to interference with laser stray light [2].

or fermion. However, we commonly encounter atoms in situations where their fermionic or bosonic nature is not manifested because the atoms are either at too high an energy or are too rarefied for quantum effects to be important. In solids, the atoms interact strongly. As a result, they are localized and don't compete for the same space so indistinguishability plays a minor role. The only material that does not become solid at low temperatures is liquid helium. When liquid helium is cooled below 2.2 K, a Bose condensate begins to form in the liquid. At these low temperatures, liquid helium behaves as a superfluid having, among other strange properties, zero viscosity.

When gaseous bosonic atoms are cold enough (so that

they have a long quantum wavelength) and dense enough (so that the spacing between the particles is on the order of the wavelength), quantum effects become important and the gas may undergo a phase transition into the Bose condensed state. This must be done in a density regime low enough so that the gas does not nucleate and form a solid. BEC was achieved with three elements: rubidium, sodium, and lithium. Although all elements (except for beryllium) have stable bosonic isotopes, the alkali atoms were used because they can be very efficiently laser-cooled. Laser cooling was the first step toward attaining BEC in these atoms.

Laser cooling works by scattering photons from moving atoms. Although the momentum of the photons is small, the alkali atoms can absorb and emit many times and therefore be quickly cooled. With the proper arrangement of laser beams, these atoms can be cooled to about  $50\mu K$  with a density of  $5 \times 10^{11} cm^{-3}$ , still far from the low temperatures and high densities needed for BEC. At this point, two other techniques are used: magnetic trapping and evaporative cooling. The atoms are trapped by creating a magnetic field minimum in free space using a simple arrangement of electromagnets. (Many atoms, including Rb, Na, and Li, are paramagnetic and can, therefore, be magnetically trapped.) Those atoms in the low field-seeking state (the magnetic moment anti-aligned with the magnetic field) can be confined in the trap. Any atom with energy above the depth of the trap can escape (evaporate). These escaping atoms necessarily have energies above the trap depth and are the hottest atoms in the gas. The atoms left in the trap cool slightly and sink deeper into the trap, thus increasing in number density. If the depth of the trap is lowered slowly, the hottest atoms are "forced" to evaporate and the sample cools continuously while its density increases. These are perfect conditions for achieving BEC! A "typical" condensate is shown in Fig.9.

## ACKNOWLEDGMENTS

I express my gratitude to Bruce Balcom for the opportunity to work on this review paper. This experience allowed me to explore and learn beyond the confines of the classroom.

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## Appendix A: Search Strategy

The searches were conducted in the **UNB World Cat catalog**, specifically employing **Scopus** database. The search results were limited to "e-book" or "Print book" written in English, as well as review articles published around the time of the first experimental realization of BEC.

To kick off my search, I compiled a list of keywords that best describe my topic and started looking for review papers on BEC in **Scopus**. Keeping track of the keywords (Table.I), I used filters and index terms within a database to refine my search. However, I noticed that most of the

papers written on BEC were overly abstract and focused primarily on specific experiments, and had little luck in finding a paper that delves into the theoretical groundings of BEC. Nevertheless, the introductory sections of the papers provided an overview of the developments in the field of ultracold atomic physics. The review paper [6] along with the nobel lecture [7] details the abckground and experimental evidence of the BEC obtained in 1995. After searching for all the relevant databases, I noticed that adding new terms yielded no new sources.

So, to gain a comprehensive theoretical background of BEC, I explored several statistical mechanics books as well as those specifically dedicated to BEC. Along the way, I realized that to fully grasp BEC, I needed a deeper understanding of Bose-Einstein statistics. Consequently, half of this review paper is devoted to elucidating Bose-Einstein statistics. In doing so, I found the following books; [4], [10], [5], [8], and [9] very helpful. While all the books describes the BEC understandably well, I found it time consuming to relate equations in different sources. Thereby, I have closely followed the equations of [4]. In addition, despite seeing similar figures in most of the sources, they often used slightly different axes titles from one another making it harder to relate to the equations used. So, one can see a better half of the figures in this paper are from the same source.

TABLE I. Search Matrix

| Research Question                                                                      |                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bose-Einstein Condensation: What is it, when does it occur, and why is it significant? |                                                                                                                                                                                                      |
| Topic                                                                                  | Bose-Einstein Statistics and Bose-Einstein Condensation                                                                                                                                              |
| Synonyms and/or related terms (keywords)                                               | Boson gas, ultracold atoms, superfluid helium <sup>4</sup> He, Superconductivity                                                                                                                     |
| Broader (more general) terms                                                           | Ultracold atomic physics, Statistical Mechanics, Quantum Statistics                                                                                                                                  |
| Narrower (more specific) terms                                                         | Bose-Einstein Condensation, Bose-Einstein Condensate, thermal properties of boson gases, Phase Diagrams                                                                                              |
| Databases used with specific queries                                                   | <b>Scopus</b> : TITLE-ABS-KEY ( Bose-Einstein Condensation ) AND PUBYEAR > 1995 AND PUBYEAR < 2024 AND ( LIMIT-TO ( DOCTYPE , "re" ) OR LIMIT-TO ( DOCTYPE , "ch" ) OR LIMIT-TO ( DOCTYPE , "bk" ) ) |